PRODUCT USER MANUAL

For

Black Sea Biogeochemical Reanalysis Product

BLKSEA_MULTIYEAR_BGC_007_005

Issue: 2.1

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	30.11.2023	all	Creation of the document	L. Vandenbulcke, M. Grégoire, R. Lecci	Copernicus Marine Product management
2.0	26.11.2024	all	Quality Improvement; Updated datasets	R. Lecci, L. Vandenbulcke, M. Grégoire	
2.1	28.05.2025	all	Merge of Reanalysis and Interim monthly datasets	R. G. de Menezes, L. Vandenbulcke, M. Grégoire, R. Lecci	Copernicus Marine Product management

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GLOSSARY AND ABBREVIATIONS

Analysis (Numerical)	a detailed study of the state of the ocean done in Near real Time based on observations and numerical model. The operational prediction centre produces 3D time-space analysis systems. A long series of analyses is of great utility for studying the behaviour of the ocean system.
BAMHBI	Biogeochemical Model for Hypoxic and Benthic Influenced areas (MAST / Université de Liège, Belgium)
BCs	Boundary Conditions
BLK	Black Sea
BFM	Biogeochemical Flux Model
CF	Climate Forecast (convention for NetCDF)
CHL	Chlorophyll
CMEMS	Copernicus Marine Environment Monitoring Service
DIC	Dissolved Inorganic Carbon
FAQ	Frequently Asked Question
Forecast (Numerical)	a computer forecast or prediction based on equations governing the motions and the forces affecting motion of fluids. The equations are based, or initialized, on specified ocean conditions at a certain place and time (NOAA Glossary).
GHER	Geo-Hydrodynamics Environmental Research laboratory of the Liege University(Belgium), and the synonym 3D circulation model
MFC	Monitoring and Forecasting Centre
NetCDF	Network Common Data Form
NOAA	National Oceanic and Atmospheric Administration

OCTAC	Ocean Color Thematic Assembly Centre
pCO2	partial pressure of carbon dioxide
PFT	Plankton Functional Types
pH	potential of Hydrogen
PU	Production Unit
ULg	Université de Liège, Belgium
3DVAR	Three-Dimensional Variational

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

I INTRODUCTION

I.1 Summary

This document is the user manual for the Copernicus Marine reanalysis product for Black Sea biogeochemistry **BLKSEA_MULTIYEAR_BGC_007_005**. An archive of reanalysis since 01/01/1992 is available on the Copernicus Marine Data Store.

The product is composed by 3D daily and monthly mean concentration of chlorophyll, nitrate, phosphate, ammonium, primary production, oxygen, phytoplankton carbon biomass, pH, dissolved inorganic carbon, alkalinity, and 2D daily and monthly mean of surface partial pressure of CO2 and surface flux of CO2.

BLKSEA_MULTIYEAR_BGC_006_005 product is organized in 21 datasets:

- 5 contain the monthly mean fields for all the variables

  cmems_mod_blk_bgc-bio_my_2.5km_P1M-m

  cmems_mod_blk_bgc-car_my_2.5km_P1M-m

  cmems_mod_blk_bgc-co2_my_2.5km_P1M-m

  cmems_mod_blk_bgc-nut_my_2.5km_P1M-m

  cmems_mod_blk_bgc-plankton_my_2.5km_P1M-m

- 5 contain the daily mean fields for all the variables

  cmems_mod_blk_bgc-bio_my_2.5km_P1D-m

  cmems_mod_blk_bgc-car_my_2.5km_P1D-m

  cmems_mod_blk_bgc-co2_my_2.5km_P1D-m

  cmems_mod_blk_bgc-nut_my_2.5km_P1D-m

  cmems_mod_blk_bgc-plankton_my_2.5km_P1D-m

- 5 contain the yearly mean fields for all the variables

  cmems_mod_blk_bgc-bio_my_2.5km_P1Y-m

  cmems_mod_blk_bgc-car_my_2.5km_P1Y-m

  cmems_mod_blk_bgc-co2_my_2.5km_P1Y-m

  cmems_mod_blk_bgc-nut_my_2.5km_P1Y-m

  cmems_mod_blk_bgc-plankton_my_2.5km_P1Y-m

- 5 contain the monthly climatology fields for all the variables.

cmems_mod_blk_bgc_bio_my_2.5km-climatology_P1M-m

cmems_mod_blk_bgc_car_my_2.5km-climatology_P1M-m

cmems_mod_blk_bgc_co2_my_2.5km-climatology_P1M-m

cmems_mod_blk_bgc_nut_my_2.5km-climatology_P1M-m

cmems_mod_blk_bgc_plankton_my_2.5km-climatology_P1M-m

-

- 1 contains the static fields for the system (coordinates, mask, and bathymetry): **cmems_mod_blk_bgc_my_2.5km_static**

The product is published on the Copernicus Marine Data Store after automatic and human quality controls. Product is available on-line and disseminated through the CMS Information System. Files downloaded are in NetCDF format.

The reanalysis system is described in the Quality Information Document (QUID)

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

I.2 History of changes

Date	Description of changes and impacted product
NOV 2023	NEW PRODUCT
NOV 2024	QUALITY IMPROVEMENT, UPDATED DATASETS
MAY 2025	MONTHLY INTERIM DATASETS MERGING

II DESCRIPTION OF THE PRODUCT SPECIFICATION

II.1 General Information

Product Line	BLKSEA_MULTIYEAR_BGC_007_005	
Geographical coverage	27.25°E - 41.10°E; 40.50°N - 47.00°N	
Variables	Nitrate Phosphate Phytoplankton Carbon Biomass Chlorophyll Net Primary Production Dissolved Oxygen (3D field and bottom value on the shelf) pH Dissolved Inorganic Carbon Alkalinity Surface partial pressure of CO2 Surface CO2 flux	
Product Type	**Reanalysis**	**Interim**
Update frequency	Twice a year	M onthly
Available time series	From 1992 to -1Y	Since the last day of Reanalysis up to the previous month
Target delivery time	NA	20th of M, at 12 UTC
Temporal resolution	cmems_mod_blk_bgc-bio_my_2.5km_P1D-m, cmems_mod_blk_bgc-car_my_2.5km_P1D-m, cmems_mod_blk_bgc-co2_my_2.5km_P1D-m, cmems_mod_blk_bgc-nut_my_2.5km_P1D-m, cmems_mod_blk_bgc-plankton_my_2.5km_P1D-m: daily mean cmems_mod_blk_bgc-bio_my_2.5km_P1M-m, cmems_mod_blk_bgc-car_my_2.5km_P1M-m, cmems_mod_blk_bgc-co2_my_2.5km_P1M-m, cmems_mod_blk_bgc-nut_my_2.5km_P1M-m,	

	cmems_mod_blk_bgc-plankton_my_2.5km_P1M-m: monthly mean cmems_mod_blk_bgc-bio_my_2.5km_P1Y-m, cmems_mod_blk_bgc-car_my_2.5km_P1Y-m, cmems_mod_blk_bgc-co2_my_2.5km_P1Y-m, cmems_mod_blk_bgc-nut_my_2.5km_P1Y-m, cmems_mod_blk_bgc-plankton_my_2.5km_P1Y-m: yearly mean cmems_mod_blk_bgc-bio_my_2.5km_ climatology_P1M-m, cmems_mod_blk_bgc-car_my_2.5km_ climatology_P1M-m, cmems_mod_blk_bgc-co2_my_2.5km_ climatology_P1M-m, cmems_mod_blk_bgc-nut_my_2.5km _climatology_P1M-m, cmems_mod_blk_bgc-plankton_my_2.5km_ climatology_P1M-m: monthly climatology
Horizontal resolution	1/40 °
Number of vertical levels	59
Format	NetCDF CF1.0
Delivery mechanisms	MDS

Table 1: BLK-MFC MultiYear product

II.2 Dataset details

BLKSEA_MULTIYEAR_BGC_007_005
Datasets: • **cmems_mod_blk_bgc-bio_my_2.5km_P1M-m,** • **cmems_mod_blk_bgc-car_my_2.5km_P1M-m,** • **cmems_mod_blk_bgc-co2_my_2.5km_P1M-m,** • **cmems_mod_blk_bgc-nut_my_2.5km_P1M-m,** • **cmems_mod_blk_bgc-plankton_my_2.5km_P1M-m** contain the monthly mean fields for all the variables • **cmems_mod_blk_bgc-bio_my_2.5km_P1D-m,** • **cmems_mod_blk_bgc-car_my_2.5km_P1D-m,**

• **cmems_mod_blk_bgc-co2_my_2.5km_P1D-m,** • **cmems_mod_blk_bgc-nut_my_2.5km_P1D-m,** • **cmems_mod_blk_bgc-plankton_my_2.5km_P1D-m** contain the daily mean fields for all the variables • **cmems_mod_blk_bgc-bio_my_2.5km_P1Y-m,** • **cmems_mod_blk_bgc-car_my_2.5km_P1Y-m,** • **cmems_mod_blk_bgc-co2_my_2.5km_P1Y-m,** • **cmems_mod_blk_bgc-nut_my_2.5km_P1Y-m,** • **cmems_mod_blk_bgc-plankton_my_2.5km_P1Y-m** contain yearly mean fields for all the variables • **cmems_mod_blk_bgc_bio_my_2.5km _climatology_P1M-m** • **cmems_mod_blk_bgc_car_my_2.5km_ climatology_P1M-m** • **cmems_mod_blk_bgc_co2_my_2.5km_ climatology_P1M-m** • **cmems_mod_blk_bgc_nut_my_2.5km_ climatology_P1M-m** • **cmems_mod_blk_bgc_plankton_my_2.5km _climatology_P1M-m** contains the monthly climatology as mean and standard deviation fields for all the variables • **cmems_mod_blk_bgc_my_2.5km_static** contains the static fields for the system: coordinates, mean sea surface level, mask, and bathymetry
Variables name in the NetCDF file and Unit: Long_name & Standard_name
no3 [mmol m-3] Nitrate mole_concentration_of_nitrate_in_sea_water
no3_avg [mmol m-3] Nitrate mole_concentration_of_nitrate_in_sea_water
no3_std [mmol m-3] Nitrate standard deviation mole_concentration_of_nitrate_in_sea_water
po4 [mmol m-3] Phosphate mole_concentration_of_phosphate_in_sea_water
po4_avg [mmol m-3] Phosphate mole_concentration_of_phosphate_in_sea_water
po4_std [mmol m-3] Phosphate standard deviation mole_concentration_of_phosphate_in_sea_water

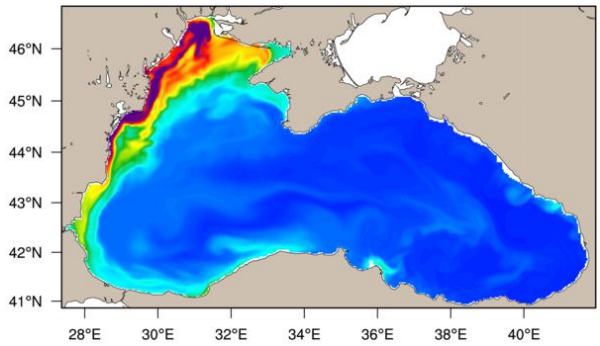
phyc [mmol m-3] Phytoplankton Carbon Biomass mole_concentration_of_phytoplankton_expressed_as_carbon_in_sea_water
phyc_avg [mmol m-3] Phytoplankton Carbon Biomass mole_concentration_of_phytoplankton_expressed_as_carbon_in_sea_water
phyc_std [mmol m-3] Phytoplankton Carbon Biomass standard deviation mole_concentration_of_phytoplankton_expressed_as_carbon_in_sea_water
chl [mg m-3] Chlorophyll mass_concentration_of_chlorophyll_a_in_sea_water
chl_avg [mg m-3] Chlorophyll mass_concentration_of_chlorophyll_a_in_sea_water
chl_std [mg m-3] Chlorophyll standard deviation mass_concentration_of_chlorophyll_a_in_sea_water
nppv [mg m-3 day-1] Net Primary Production net_primary_production_of_biomass_expressed_as_carbon_per_unit_volume_in_sea_water
nppv_avg [mg m-3 day-1] Net Primary Production net_primary_production_of_biomass_expressed_as_carbon_per_unit_volume_in_sea_water
nppv_std [mg m-3 day-1] Net Primary Production standard deviation net_primary_production_of_biomass_expressed_as_carbon_per_unit_volume_in_sea_water
o2 [mmol m-3] Dissolved oxygen mole_concentration_of_dissolved_molecular_oxygen_in_sea_water
o2_avg [mmol m-3] Dissolved oxygen mole_concentration_of_dissolved_molecular_oxygen_in_sea_water
o2_std [mmol m-3] Dissolved oxygen standard deviation mole_concentration_of_dissolved_molecular_oxygen_in_sea_water
o2b [mmol m-3] Dissolved oxygen on the shelf bottom mole_concentration_of_dissolved_molecular_oxygen_in_sea_water
o2b_avg [mmol m-3] Dissolved oxygen on the shelf bottom mole_concentration_of_dissolved_molecular_oxygen_in_sea_water
o2b_std [mmol m-3] Dissolved oxygen standard deviation on the shelf bottom mole_concentration_of_dissolved_molecular_oxygen_in_sea_water

ph [1] Ocean pH sea_water_ph_reported_on_total_scale
ph_avg [1] Ocean pH sea_water_ph_reported_on_total_scale
ph_std [1] Ocean pH standard deviation sea_water_ph_reported_on_total_scale
dissic [mol m-3] Dissolved Inorganic Carbon mole_concentration_of_dissolved_inorganic_carbon_in_sea_water
dissic_avg [mol m-3] Dissolved Inorganic Carbon mole_concentration_of_dissolved_inorganic_carbon_in_sea_water
dissic_std [mol m-3] Dissolved Inorganic Carbon standard deviation mole_concentration_of_dissolved_inorganic_carbon_in_sea_water
talk [mol m-3] Alkalinity sea_water_alkalinity_expressed_as_mole_equivalent
talk_avg [mol m-3] Alkalinity sea_water_alkalinity_expressed_as_mole_equivalent
talk_std [mol m-3] Alkalinity standard deviation sea_water_alkalinity_expressed_as_mole_equivalent
spco2 [Pa] Surface partial pressure of CO2 surface_partial_pressure_of_carbon_dioxide_in_sea_water
spco2_avg [Pa] Surface partial pressure of CO2 surface_partial_pressure_of_carbon_dioxide_in_sea_water
spco2_std [Pa] Surface partial pressure of CO2 standard deviation surface_partial_pressure_of_carbon_dioxide_in_sea_water
fpco2 [kg m-2 s-1] Surface CO2 flux surface_downward_mass_flux_of_carbon_dioxide_expressed_as_carbon
fpco2_avg [kg m-2 s-1] Surface CO2 flux surface_downward_mass_flux_of_carbon_dioxide_expressed_as_carbon
fpco2_std [kg m-2 s-1] Surface CO2 flux standard deviation surface_downward_mass_flux_of_carbon_dioxide_expressed_as_carbon
e1t [m] Cell dimension along X axis
e2t [m]

Cell dimension along Y axis
e3t [m] Cell dimension along Z axis cell_thickness
mask [1] Land-sea mask: 1 = sea ; 0 = land sea_binary_mask
deptho [m] Bathymetry sea_floor_depth_below_geoid
deptho_lev [1] Model level number at sea floor model_level_number_at_sea_floor

Table 2: list of the datasets and variable names and unit for the BLKSEA_MULTIYEAR_BGC_007_005 product

II.3 Production System Description

Domain	BLKSEA (27.25°E - 41.10°E; 40.50°N - 47.00°N)
Resolution and grid	1/40°; regular grid; 591 x 261
Geographic coverage	This product covers the Black Sea, the horizontal resolution is 1/40 (approx 2.5 km), the vertical grid is composed of 59 unevenly spaced levels. 
Model Version **Production unit** **Period of validation**	▪ nemo_bamhbi ▪ MARINES , Belgium ▪ 1992 -2021

Vertical grid	The product is delivered at the model native grid, with up to 59 vertical levels: levels [m]: 0.25, 0.77, 1.32, 1.89, 2.49, 3.12, 3.80, 4.54, 5.33, 6.20, 7.15, 8.20, 9.37, 10.68, 12.16, 13.82, 15.72, 17.89, 20.38, 23.23, 26.53, 30.34, 34.76, 39.88, 45.83, 52.75, 60.79, 70.14, 81.01, 93.63, 108.27, 125.23, 144.83, 167.43, 193.41, 223.17, 257.14, 295.75, 339.40, 388.52, 443.45, 504.52, 571.99, 646.03, 726.73, 814.09, 908.02, 1008.35, 1114.81, 1227.10, 1344.84, 1467.64, 1595.06, 1726.70, 1862.12, 2000.92, 2142.72, 2287.16, 2433.67
Atmospheric forcing	ECMWF ERA5 reanalysis fields (1 hour, 0.25 deg. horizontal resolution)
Boundary forcing	The ocean model NEMO 4.2 is using conservation laws at the Bosphorus open boundary. The barotropic flux is computed based on the sea surface elevation at the Bosphorus mouth; the temperature and salinity of the incoming temperature follow respectively a zero-gradient approach and the salt-conservation law. In NEMO-BAMHBI, the biogeochemical tracer concentrations in the incoming water are either specified or follow a zero-gradient approach.
Assimilation scheme	fixed-base SEEK
Assimilated observations	surface chlorophyll
Initial conditions (spin-up process)	Physical fields: NEMO-BAMHBI model outputs (previous version) Biogeochemistry: NEMO-BAMHBI model outputs (previous version) Carbonate component: Proprietary climatology built from observations
Bathymetry	GEBCO 30sec interpolated on the model grid
River run-off	Danube: data from the Romanian Hydrological institute. Other rivers: SESAME / PERSEUS deliverable

The biogeochemical reanalyses for the Black Sea are produced by the MARINES / Belgium Production Unit by means of the NEMO 4.2 circulation model online coupled with the BAMHBI biogeochemical model. The workflow runs on the ULiege / CECI supercomputing facility "nic5". Multi-year product is produced yearly , updated monthly with Interim data .

II.4 Grid

The product is delivered on the native NEMO-BAMHBI grid with a horizontal resolution of approx. 2.5 km, and 59 unevenly-spaced z-levels.

II.5 Vertical Levels

The vertical layers are unevenly spaced; the mid-cell depth is given in section II.3

II.6 Processing information

II.6.1 Update Time

The Black Sea Biogeochemical interim datasets are updated monthly on the 20th (addition of the previous month).

II.6.2 Temporal extent of reanalysis stored on delivery mechanism

The product is available from 01/01/1992 .

II.6.3 Time averaging

For the monthly dataset, the fields are monthly averages over the calendar month (first to last day of the month). For the daily dataset, the fields are daily averages over a day (00:00 to 24:00 UTC, centered at midday).

III FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:

What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:

how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

blksea_multiyear_bgc_007_005 files nomenclature is based as follows:

{valid date}_{freq flag}-{producer}--{parameter}-{config}-{region}-{bul date}_{product type}-sv{file version}.nc

where

- ✓ **valid date** YYYYMMDD is the validity day of the data in the file
- ✓ **freq flag** is the frequency of data values in the file (h = hourly, d = daily, m=monthly, y=yearly)
- ✓ **producer** is a short version of the CMS production unit
- ✓ **config** identifies the producing system and configuration
- ✓ **region** is a maximum six letter code for the region
- ✓ **parameter** is a four-letter code for the parameter or parameter set from Standard BODC.
- ✓ **bul date** bYYYYMMDD is the bulletin date the product was produced
- ✓ **product type** is a two-letter code for the product type, for example fc for forecast, an for analysis and sm for hindcast.
- ✓ **file version** is xx.yy where xx is the CMS version and yy is an incremental version number

Table 1 shows the nomenclature for the **BLKSEA_MULTIYEAR_BGC_007_005** products.

Table 1 Description of the nomenclature for BLKSEA_MULTIYEAR_BGC_007_005

valid date	YYYYMMDD
freq flag	m (monthly) d (daily)
producer	ULg
config	nemo_bamhbi
region	BLK
parameter	NUTR PFTC BIOL CARB CO2F

bul date	bYYYYMMDD
product type	re (reanalysis) in (interim)
file version	02.00

Example for a reanalysis file:

20140501_d-ULg--PFTC-nemo_bamhbi-BS-b20210112_re-sv02.00.nc

This is the daily mean field of biogeochemistry centered at 12:00 UTC of the 1st May 2014, and the time coverage is from midnight (00:00 UTC) of the 1st May 2014 to midnight (00:00 UTC) of the 2nd May 2014 .

20140501_m-ULg--PFTC-nemo_bamhbi-BS-b20210112_re-sv02.00.nc

This is the monthly mean field of biogeochemistry for May 2014. The mean is computed from midnight (00:00 UTC) of the 1st May 2014 to midnight (00:00 UTC) of the 1st June 2014

20140101_y-ULg--PFTC-nemo_bamhbi-BS-b20210112_re-sv02.00.nc

This is the yearly mean field of biogeochemistry for 2014. The mean is computed from midnight (00:00 UTC) of the 1st Jan 2014 to midnight (00:00 UTC) of the 1st Jan 2015.

This is the monthly mean field of biogeochemistry for the month of March 2020. The mean is computed from midnight (00:00 UTC) of the 1st March 2020 to midnight (00:00 UTC) of 1st April 2020.

IV.1 Other information: mean centre of Products, land mask value, missing value

The missing value for this product is: 1.e+20

Land mask is equal to "_FillValue" (see variable attribute on NetCDF file).

Real_Value = (Display_Value X scale_factor) + add_offset

The BGC reanalyses are using the NetCDF4 format without offset nor scale factors.

IV.2 File size

DATASET NAME	FILE NAME	DIMENSION [MB]
cmems_mod_blk_bgc-nut_my_2.5km_P1D-m	{date1}_d-ULg--NUTR-nemo_bamhbi-BS-b{date2}_re-sv02.00.nc	7.5-10.0
cmems_mod_blk_bgc-plankton_my_2.5km_P1D-m	{date1}_d-ULg--PFTC-nemo_bamhbi-BS-b{date2}_re-sv02.00.nc	7.5-10
cmems_mod_blk_bgc-bio_my_2.5km_P1D-m	{date1}_d-ULg--BIOL-nemo_bamhbi-BS-b{date2}_re-sv02.00.nc	8.2-11.5
cmems_mod_blk_bgc-car_my_2.5km_P1D-m	{date1}_d-ULg--CARB-nemo_bamhbi-BS-b{date2}_re-sv02.00.nc	9.0-12.0
cmems_mod_blk_bgc-co2_my_2.5km_P1D-m	{date1}_d-ULg--CO2F-nemo_bamhbi-BS-b{date2}_re-sv02.00.nc	0.3-1.5

cmems_mod_blk_bgc-nut_my_2.5km_P1M-m	{date1}_m-ULg--NUTR-nemo_bamhbi-BS-b{date2}_re-sv02.00.nc	7.5-10.0
cmems_mod_blk_bgc-plankton_my_2.5km_P1M-m	{date1}_m-ULg--PFTC-nemo_bamhbi-BS-b{date2}_re-sv02.00.nc	7.5-10.0
cmems_mod_blk_bgc-bio_my_2.5km_P1M-m	{date1}_m-ULg--BIOL-nemo_bamhbi-BS-b{date2}_re-sv02.00.nc	8.2-11.5
cmems_mod_blk_bgc-car_my_2.5km_P1M-m	{date1}_m-ULg--CARB-nemo_bamhbi-BS-b{date2}_re-sv02.00.nc	9.0-12.0
cmems_mod_blk_bgc-co2_my_2.5km_P1M-m	{date1}_m-ULg--CO2F-nemo_bamhbi-BS-b{date2}_re-sv02.00.nc	0.3-1.5
cmems_mod_blk_bgc-bio_my_2.5km_P1Y-m	{date1}_y-ULg--BIOL-nemo_bamhbi-BS-b{date2}_in-sv02.00.nc	8.2-11.5
cmems_mod_blk_bgc-nut_my_2.5km_P1Y-m	{date1}_y-ULg--NUTR-nemo_bamhbi-BS-b{date2}_in-sv02.00.nc	7.5-10.0
cmems_mod_blk_bgc-car_my_2.5km_P1Y-m	{date1}_y-ULg—CARB-nemo_bamhbi-BS-b{date2}_in-sv02.00.nc	9.0-12.0
cmems_mod_blk_bgc-co2_my_2.5km_P1Y-m	{date1}_y-ULg--CO2F-nemo_bamhbi-BS-b{date2}_in-sv02.00.nc	0.3-1.5
cmems_mod_blk_bgc-plankton_my_2.5km_P1Y-m	{date1}_y-ULg--PFTC-nemo_bamhbi-BS-b{date2}_in-sv02.00.nc	7.5-10.0
cmems_mod_blk_bgc_bio_my_2.5km_ climatology_P1M-m	ULg--BIOL- nemo_bamhbi-BS_mean_1992_2022_ {month } .nc ULg--BIOL-nemo_bamhbi-BS_ stddev_1992_2022_{month}.nc	15 17
cmems_mod_blk_bgc_nut_my_2.5km_ climatology_P1M-m	ULg--NUTR- nemo_bamhbi-BS_mean_1992_2022_ {month } .nc ULg--NUTR-nemo_bamhbi-BS_ stddev_1992_2022_{month}.nc	14 15
cmems_mod_blk_bgc_car_my_2.5km_ climatology_P1M-m	ULg--CARB- nemo_bamhbi-BS_mean _1992_2022_ {month } .nc ULg--CARB-nemo_bamhbi-BS_ stddev_1992_2022_{month}.nc	17 25

cmems_mod_blk_bgc_co2_my_2.5km_ climatology_P1M-m	ULg--CO2F- nemo_bamhbi-BS_mean_1992_2022_ {month } .nc ULg--CO2F-nemo_bamhbi-BS_ stddev_1992_2022_{month}.nc	0.4
cmems_mod_blk_bgc_plankton_ my_2.5km_ climatology_P1M-m	ULg--PFTC- nemo_bamhbi-BS_mean_1992_2022_ {month } .nc ULg--PFTC-nemo_bamhbi-BS_ stddev_1992_2022_{month}.nc	18
cmems_mod_blk_bgc_my_2.5km_ static	BS-MFC_007_005_coordinates .nc BS-MFC_007_005_mask_bathy.nc	38 11

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